



4425 Manchester Rd

Kalamazoo, MI 49001

Phone 269 381-9666

Fax 269 381-9698

www.karlabs.com

**City of Cadillac, Water Resources
200 North Lake Street
Cadillac, MI 49601**

Attn : Ms. Amy Vail

KAR Project No. : 612234

Date Reported : 09/27/16

Date Activated : 09/13/16

Date Due : 09/27/16

Date Validated : 09/27/16

Project

Description : Analysis of one sludge sample for Fall Biosolids.

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received. If a sample was identified as drinking water under the Safe Drinking Water Act, the "Comments" column may also contain federal drinking water information including MCL which is the Maximum Contaminant Level set by USEPA. Values enclosed in brackets ([]) are Secondary MCL's and are non-enforceable guidelines for aesthetic quality.

If you wish to contact us about this work please mention KAR Project No. 612234. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

A handwritten signature in black ink, appearing to read "David R. Alkema".

David R. Alkema
Laboratory Manager

KAR Laboratories, Inc. maintains Full Certification status for Bacteriology, Inorganics, Regulated Organics and Synthetic Organics through USEPA, Michigan Department of Public Health and Indiana State Department of Health. This report may only be reproduced in full and not without the written consent of **City of Cadillac, Water Resources**.

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **612234**

Date Reported : **09/27/16**

Attest: 
David R. Alkema, Lab Manager

Project

Description: *Analysis of one sludge sample for Fall Biosolids.*

Sample ID : **"Digester #4"**

Sampled By : *TL of City of Cadillac, Water Resources*

Date Received : **09/13/16**

Sample Date : **09/12/16**

Sample Type : **sludge**

Sample Time : **1330**

KAR Sample No. : **612234-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, Hg	Completed		EPA 7471A	09/26/16	MID	
Prep, metals	Completed		EPA 200.7/8	09/16/16	MID	
Aluminum, total	2210	mg/kg total solids	EPA 6010B	09/23/16	JHB	Matrix effect observed; result is approximate. Detection limit = 77
Arsenic, total	15.5	mg/kg total solids	EPA 200.8	09/26/16	NHM	Matrix effect observed; result is approximate. Detection limit = 7.7
Cadmium, total	<3.8	mg/kg total solids	EPA 200.8	09/26/16	NHM	Matrix effect observed; result is approximate.
Calcium, total	16,000	mg/kg total solids	EPA 6010B	09/23/16	JHB	Matrix effect observed; result is approximate. Detection limit = 77
Chromium, total	50.5	mg/kg total solids	EPA 200.8	09/26/16	NHM	Matrix effect observed; result is approximate. Detection limit = 7.7
Copper, total	467	mg/kg total solids	EPA 6010B	09/23/16	JHB	Matrix effect observed; result is approximate. Detection limit = 16
Lead, total, by ICP	<39	mg/kg total solids	EPA 6010B	09/23/16	JHB	
Magnesium, total	2740	mg/kg total solids	EPA 6010B	09/23/16	JHB	Matrix effect observed; result is approximate. Detection limit = 77
Mercury, total	0.56	mg/kg total solids	EPA 7471A	09/26/16	MID	Matrix effect observed; result is approximate. Detection limit = 0.09
Molybdenum, total	20.2	mg/kg total solids	EPA 200.8	09/26/16	NHM	Matrix effect observed; result is approximate. Detection limit = 15
Nickel, total	<15	mg/kg total solids	EPA 200.8	09/26/16	NHM	Matrix effect observed; result is approximate.
Potassium, total	1740	mg/kg total solids	EPA 6010B	09/23/16	JHB	Matrix effect observed; result is approximate. Detection limit = 77
Selenium, total	<3.8	mg/kg total solids	EPA 200.8	09/26/16	NHM	Matrix effect observed; result is approximate.
Silver, total	1.5	mg/kg total solids	EPA 6010B	09/20/16	JHB	Detection limit = 1.0
Sodium, total	2550	mg/kg total solids	EPA 6010B	09/23/16	JHB	Matrix effect observed; result is approximate. Detection limit = 770
Zinc, total	828	mg/kg total solids	EPA 6010B	09/23/16	JHB	Matrix effect observed; result is approximate. Detection limit = 16
Chloride	4840	mg/kg total solids	EPA 300.0A	09/26/16	ALK	Detection limit = 160
Nitrogen, ammonia	37,000	mg/kg total solids	EPA 350.1	09/20/16	JWW	Detection limit = 1300
Nitrogen, nitrate	<3.1	mg/kg total solids	EPA 353.2	09/26/16	AJK	
Nitrogen, total kjeldahl	90,200	mg/kg total solids	EPA 351.2	09/14/16	ALK	Detection limit = 6900
Phosphorus, total (as P)	45,000	mg/kg total solids	SM 4500-P B5,E	09/20/16	EIF	Detection limit = 6200
Sulfate	63,300	mg/kg total solids	EPA 300.0A	09/26/16	ALK	Detection limit = 1600
Waste pH	7.7	S.U.	EPA 9040C	09/19/16	AJK	
Solids, total	3.24	% by weight	EPA 160.3	09/15/16	JWW	Detection limit = 0.01
Solids, volatile	53.70	% of total solids	EPA 160.4	09/15/16	JWW	Detection limit = 0.01

www.karlabs.com

[illegible]